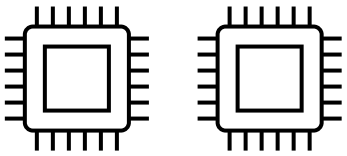
NOVA SCHOOL OF SCIENCE & TECHNOLOGY

*Eletrónica para Micro-Sistemas / Electronics for Micro-Systems*

# Lab #1

# Analog Front-End for Resistive Sensors

## NTC Temperature Sensor Measurement and Soil-Moisture Extension



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# 1- Introduction

This lab experiment intends to introduce the design and experimental validation of an analog front-end (AFE) for a resistive sensor. The mandatory sensor is a 10 kΩ NTC thermistor used for air-temperature measurement. The acquired signal is processed by a Raspberry Pi Pico 2 W and used to actuate a relay.

A firmware template for the Pico 2 W and an almost complete PC graphical application are provided so that laboratory time is concentrated on electronics, measurement and system integration while making a first entry on MCU programming.

The lab is mainly focused on the NTC thermistor. In addition, an extra and optional challenge is proposed involving the use of soil moisture sensor.

*Workload*

The work plan includes two 2-hour (total of 4 hours) laboratory sessions. Preparation before the first session is highly recommended. The lab report is also intended to be short (4 pages plus anexes) and technically focused.

| Stage | Main activity | Target |
|---|---|---|
| **Before Session 1** | NTC model, preliminary AFE design and basic LTspice check | Arrive ready to build |
| **Session 1 - 2 h** | Build and test the NTC AFE; connect Pico; obtain live temperature | Working sensing chain |
| **Session 2 - 2 h** | Characterize the system; add relay actuation; optional soil sensor | Complete sensing-and-actuation system |

Grading scheme

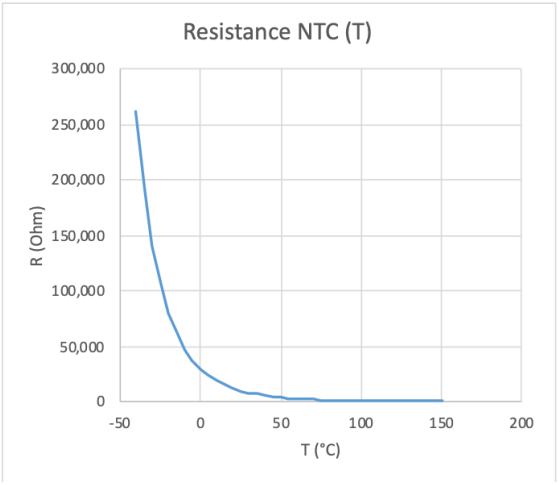
- Up to 18: NTC sensor and relay
- Up to 20: mandatory NTC sensor and relay, plus the integration of the soil moisture sensor.

# 2- Sensors

## 2.1 – Temperature Sensor: NTC thermistor

A Negative Temperature Coefficient (NTC[1]) thermistor is a sensor whose resistance suffers a decrease when a temperature rise occurs.




a) b)

*Figure 1 - NTC thermistor characteristic curve.*

The relation between resistance and temperature is non-linear and can be modeled to a high degree of accuracy using some modeling techniques. The first one, known as "Beta model", the resistance is given by

$$R = R_0 \times e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)} \quad (1)$$

where $R_0$ is resistance at temperature $T_0$ (in Kelvin), and ß is the Beta coefficient associated with a particular NTC. This equation is useful over a narrow temperature range, with an accuracy of approximately +- 1oC over the temperature range of 0°C to 100 °C.

A more accurate model is described by the Steinhart-Hart equation (https://en.wikipedia.org/wiki/Steinhart%E2%80%93Hart_equation) ,

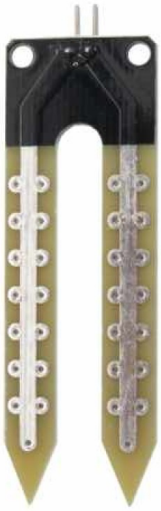
$$\frac{1}{T} = A + B.\ln(R) + C.[\ln(R)]^3 \quad (2)$$

where T is the temperature in kelvin, and R is resistance in ohms at temperature T. The Steinhart-Hart coefficients A, B and C depends on the type, model of the thermistor, and temperature range of interest.

[1] https://en.wikipedia.org/wiki/Thermistor

The thermistor used in the laboratory has a nominal resistance of 10 kΩ at its reference temperature. Confirm the exact model available at the bench and obtain its parameters from the corresponding datasheet.

## 2.2 – Resistance based soil moisture sensor



The available soil-moisture sensor is resistive, presenting a minimum resistance when placed in fully water-saturated soil. Check the characteristics of the sensor provided and determine whether it needs to be connected in a voltage-divider configuration or whether it already provides a voltage that can be used directly. In fact, some soil-moisture sensors already include internal signal-conditioning circuitry and provide an analog output voltage directly, typically in the range of 0–3.3 V or 0–5 V. In this case, an external voltage divider is not required. Other sensors are purely resistive and therefore require an external voltage divider. Before assembling any circuit, inspect the sensor and use a multimeter to determine whether it has only two terminals, indicating a purely resistive sensor, or three or four terminals, which usually indicates that additional circuitry is integrated. Consult the teaching staff if in doubt.

Because the relationship between the sensor resistance and the actual soil water content is strongly dependent on the soil type, probe geometry, compaction, conductivity, and other factors, it is difficult to use a simple analytical expression to convert the sensor reading into an absolute moisture value.
For this laboratory, an empirical calibration approach will therefore be used. Measure the sensor output under two reference conditions:

- dry soil, used as the lower-moisture reference;
- water-saturated soil, used as the upper-moisture reference.

The measured values can then be used to define a relative moisture scale between these two conditions. For example, the sensor output may be normalized to a range from 0% (dry reference) to 100% (water-saturated reference).
This percentage should be interpreted only as a relative moisture indication for the tested setup, not as an absolute measurement of the volumetric water content of the soil.

# 3- System design and testing

## 3.1 – Design of the temperature meter and relay system

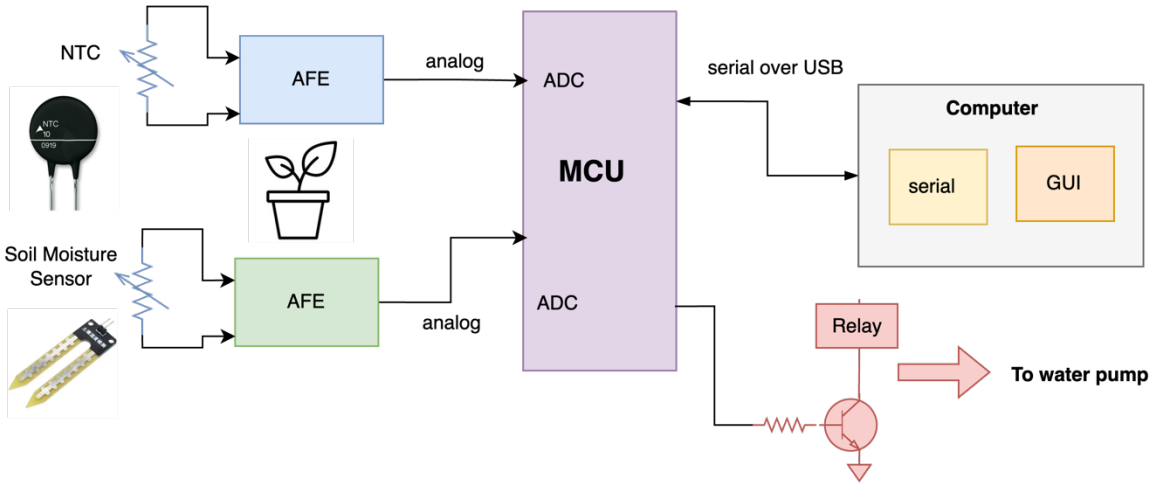



*Figure 2 - Temperature sensing system with 3 or 4 types of temperature sensors.*

The block diagram of the system that you are required to design is represented in figure 2. The main design focus is AFE for the NTC sensor and the complete chain is to be implement and fully tested.

The first sensor is an NTC thermistor which needs an Analog Front-End (AFE) circuit to prepare the sensor signal to be digitized by the microcontroller ADC. This AFE must be designed and therefore a circuit schematic must be proposed, designed, implemented, and tested. As mentioned before, the signal from the AFE is digitalized by the MCU ADC. The ADC digital output is processed (in software) to calculate the corresponding temperature value using the Beta model equation (1), for example.

In summary, the design/development work consists of:

### Sensor and AFE

- The thermistor is a 10kΩ NTC. Obtain parameter ß from the manufacturer data. Three models are available in the LAB. The datasheets are available on the course page @moodle. Please confirm the model with the teaching staff.
- Select a useful air-temperature range for the experiment and determine the corresponding NTC resistance range.
- Avoid unnecessary NTC current and consider possible self-heating.
- Consider a single supply design. Please, remember that this AFE is intended to connect to an ADC which has a limited voltage input range
- When selecting the OPAMP, please consider the gain bandwidth product, slew rate, common mode input range, input offset voltage, and input current biasing. Please indicate your conclusions in the report.
- The total AFE power dissipation should be indicated in the report.

- Estimate the main sources of error introduced by the AFE.
- From the analysis of the AFE, include in your report:
    - Theoretical transfer characteristic of the AFE
    - Impact of the offset of the OPAMP in the transfer function of the AFE
    - Any other aspects that you consider important to achieve a good design.

## Actuator and AFE

The actuator is basically 5V DC / 220V AC relay that is controlled by a MCU output digital pin. Use the measured temperature to control a relay. A relay-driver circuit based on a transistor and a flyback is needed (why?). Why we need to program a control characteristic with hysteresis?

## Components

1. Available OPAMPS are:
    a. MCP6004, from MicroChip. https://www.microchip.com/en-us/product/MCP6004.
    b. LM324, from TI. https://www.ti.com/product/LM324?keyMatch=LM324
    c. TL084, from TI, https://www.ti.com/product/TL084#product-details

    When selecting an OPAMP for the application, check if the gain-bandwidth product is high enough, slew-rate, voltage offset, and current offset are low, linearity, etc.

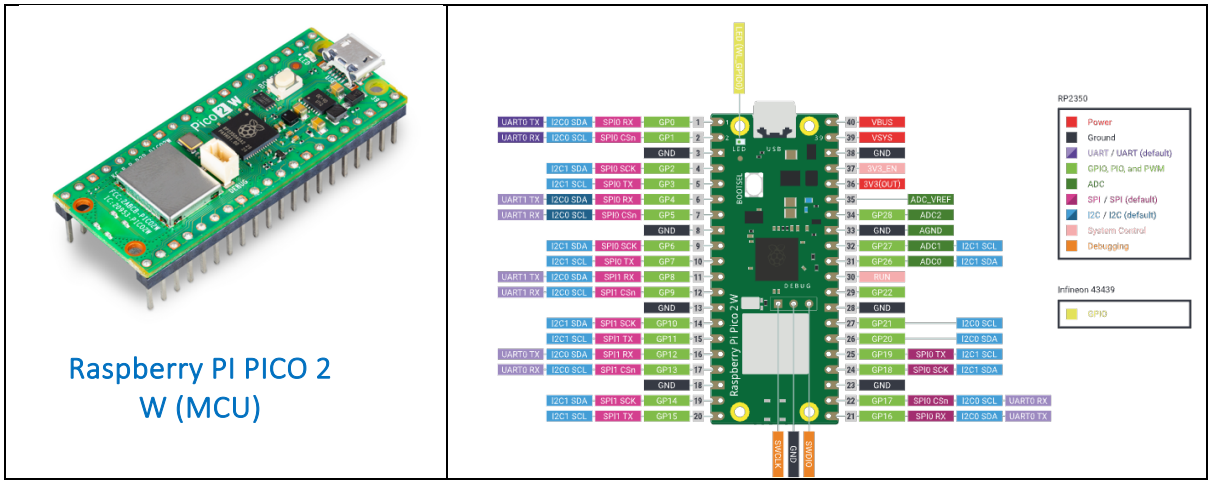
2. Resistors: consider the use of resistors from series E12, with a maximum tolerance of 5%.

3. Capacitors: consider the use of capacitors from series E12, with a 5% of tolerance

4. Temperature sensors: NTC (10kΩ) Datasheets are available on moodle.

5. Consider the following microcontroller (MCU), see Figure 4:

    - RPI Pico 2: *https://www.raspberrypi.com/documentation/microcontrollers/pico-series.html*; The microcontroller board can be programmed using the same Arduino IDE and share a similar program structure based on c/c++ language. Alternative IDE: Visual Studio Code ( https://code.visualstudio.com/ )




https://www.raspberrypi.com/documentation/microcontrollers/pico-series.html

*Figure 3 - RPI Pico 2 W microcontroller board.*

### Firmware and GUI software

The C/C++ firmware for the RPI Pico 2W MCU template is provided. ADC initialization, the main program structure and serial communication are already included. The group must complete the calculations that depend on its own AFE.

```
adc_code = read_adc();          // provided
v_adc    = ...;                   // complete
r_ntc    = ...;                   // complete using your AFE equation
temp_C   = ...;                   // complete using the Beta model
send_data(...);                   // provided
```

Despite the given code, it is higlhy recommended/required that the code is analysed and find possible bugs. Questions will be raised by the teaching staff during the experiments and the answers resutls will be registered and considered during the classification process.

A Python graphical application (GUI) is also be provided. It already contains the serial-port handling and the main GUI. Only minor adaptation should be necessary to match the data generated by the group. The GUI is a visualization tool for this laboratory and is not a major assessment component. The GUI is based on Tkinker library (https://realpython.com/python-gui-tkinter/ )

## 3.2 – Simulation

Considering the design and component sizing for the NTC case, simulate the AFE in LTspice. Obtain the transfer characteristic of the AFE with respect to the variation of the NTC resistance (related with temperature). However, a report containing extra analysis like monte-carlo simulation or other type of simulations (octave modeling or python modeling) will score higher.

## 3.3 – Implementation and experimental testing

The AFE and RPI Pico 2 setup will be implemented on a breadboard and tested in the lab. Due to the limited time, it is strongly recommended that the wire connections on the breadboard are well and carefully planned before the class.

This is "DC" (low-frequency) test-type experiment and therefore is straightforward. The main challenge is related to the difficulty to change the temperature of the thermistor in a controlled way. An electronic thermometer will be available in the Lab which can be used to obtain approximate room temperature values, used as a reference. Try to change the temperature of the sensor and collect the data from the RPI Pico 2 and also from a multimeter connected to the output of the AFE. This data will be used during the post-processing phase after the experiment. The results of this post-processing must be included in the report.

Calibration is needed. However, do not spend too much laboratory time trying to obtain high precision temperature calibration. The available temperature meter is a commercial consumer device and is used only as an approximate reference.

Suggested targets for the two lab sessions:

1. Session 1 target: before leaving the laboratory, the group should have a working NTC acquisition chain and should be able to observe a temperature change on the PC when the NTC is warmed.
2. Session 2 target: validate the electrical behavior of the AFE and complete the sensing-and-actuation chain by controlling a relay from the measured temperature.
    a. Obtain a small set of measurements under different temperature conditions. The purpose is to check consistency between the sensor model, the AFE design, the simulation and the measured electrical quantities. The commercial temperature meter is not a calibration instrument. Therefore, do not claim high absolute accuracy. Use it only to check that the complete system reports a plausible temperature.
    b. Relay control: Use the measured temperature to control a relay. A relay-driver circuit based on a transistor and a flyback diode will be provided or discussed in class; the group must connect it correctly and verify its operation. Note: imagine that with this relay we are able of controlling the heating temperature of the water in an electric kettle.

## 3.4 – Challenge

After the mandatory NTC system is operating correctly, groups may add the resistive soil-moisture sensor using a second ADC channel. This part is less guided and is intended to test whether the design method can be transferred to another resistive sensor.

Consider the following list of remarks and suggestions:

- Measure the sensor resistance in at least two clearly different moisture conditions.

- Determine a useful resistance range and design an interface compatible with the Pico ADC.
- Simulate and implement the interface.
- Acquire the NTC and soil-moisture channels in the same firmware.
- Adapt the provided GUI so that temperature and soil-moisture information are shown together. (code is already prepared, only need activate a flag)
- Define a relative moisture indication using experimentally obtained dry and wet reference conditions.
- A 0–100% indication may be used as a relative scale between the selected dry and wet reference points. It must not be presented as an absolute soil-water-content measurement unless a suitable calibration procedure has been performed.

## 4- Evaluation and Report

The evaluation of this lab is determined by the level of participation of each group element and a report describing the design, implementation, and testing of the designed system. Report and lab work in group (max 3 students).

The experimental Lab will occur in person, the report due date is indicated in Moodle. The report (PDF format, not in docx or Latex) must be uploaded in the Moodle platform. All netlists, python scripts, excel files, or other files shall be uploaded.

Report length: **maximum 4 pages** of technical content, excluding the cover page, table of contents, references and appendices. Groups completing the optional soil-moisture challenge may use one additional page.

Keep the report concise and focused on the design, simulation, experimental results and discussion.

Grading scheme

- Up to 18: NTC sensor and relay
- Up to 20: mandatory NTC sensor and relay, plus the integration of the soil moisture sensor